Development of Autonomous Failure Maintenance System for Semiconductor Manufacturing

Nuri Han
Samsung Electronics
Seoul, Repulic of Korea
n.han@samsung.com

Jiwon Seo
Samsung Electronics
Seoul, Repulic of Korea
jiwon23.seo@samsung.com

Jonghee Ha
Samsung Electronics
Seoul, Repulic of Korea
jong2.ha@samsung.com

Jihyung Oh
Samsung Electronics
Seoul, Repulic of Korea
lifehappy.oh@samsung.com

Jinwoo Lee
Samsung Electronics
Seoul, Repulic of Korea
jinwoo93.lee@samsung.com

Boram Jeong
Samsung Electronics
Seoul, Repulic of Korea
boram.jeong@samsung.com

Jongbin Park
Samsung Electronics
Seoul, Repulic of Korea
jongbin.park@samsung.com

Gilhwan Kim
Samsung Electronics
Seoul, Repulic of Korea
gilhwan1.kim@samsung.com

Yohwan Joo
Samsung Electronics
Seoul, Repulic of Korea
yohwan.joo@samsung.com

Abstract

Semiconductor equipment failure analysis is critical in the fast-paced semiconductor industry, where complex and sensitive components are susceptible to failures that can lower productivity, raise costs, and shorten equipment lifespan. But conventional failure analysis methods rely on time series-based fault detection classification to identify the cause of failure, and have limitations in relying on experts for appropriate actions. We present a fully deployed autonomous maintenance system that closes the failure recovery loop, from detecting anomalies to executing corrective actions, by integrating graph-based log analysis, LLM-based semantic reasoning and hybrid retrieval mechanisms. Deployed across 762 photolithography equipment at Samsung Electronics fab from 2021 to 2025, the system handled over 85,000 real-world breakdowns, reduced Mean Time to Recovery by 7 minutes on average. This autonomous system enhances productivity, reduces costs, and maintains high product quality, supporting the transition to fully automated environments in the semiconductor industry.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → **Natural language processing**.

Keywords

Domain-Adapted Retrieval Augmented Generation;Semiconductor Fab Automation;Large Language Models;Failure Diagnosis

ACM Reference Format:
Nuri Han, Jiwon Seo, Jonghee Ha, Jihyung Oh, Jinwoo Lee, Boram Jeong, Jongbin Park, Gilhwan Kim, and Yohwan Joo. 2025. Development of Autonomous Failure Maintenance System for Semiconductor Manufacturing. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3746252.3761556

This work is licensed under a Creative Commons Attribution 4.0 International License.
CIKM '25, Seoul, Republic of Korea
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2040-6/2025/11
https://doi.org/10.1145/3746252.3761556

1 Introduction

Semiconductor manufacturing requires high precision and uninterrupted operation across complex equipment. Equipment failures in fabs can cause production delays, yield loss, and costly downtime. Rapid and accurate recovery is therefore essential for maintaining productivity.

Traditionally, engineers manually inspect event logs and sensor data to infer failure causes and determine recovery actions. These methods rely heavily on expert intuition and static rules, making them slow to adapt to unfamiliar or compound failures. Moreover, event logs are often underutilized due to their unstructured format and semantic ambiguity.

To address these limitations, we propose an autonomous failure maintenance system for semiconductor fabs. It integrates graph-based event propagation analysis, LLM-based semantic scoring, domain-adapted retrieval-augmented generation (RAG), and hybrid prompting to automate failure diagnosis and recovery.

Figure 1 shows the deployed architecture. Failures are detected from live logs, analyzed through centrality-based graphs and LLM scoring (Section 3.1), and reformulated as queries for action planning. Recovery actions are synthesized using symbolic SQL retrieval and HyDE + Least-to-Most prompting (Section 3.2), then either executed automatically or reviewed by engineers (Section 3.3). Engineer feedback and historical documents are reflected to update the knowledge base (Section 3.4).

The system has been deployed on 762 photolithography equipment at Samsung Electronics fab since 2021, processing over 85,000 breakdowns and autonomously resolving over 70% within one minute. By combining language reasoning with execution control, it significantly reduces downtime and supports continuous learning in high-stakes manufacturing environments.

The key contributions of this work are:

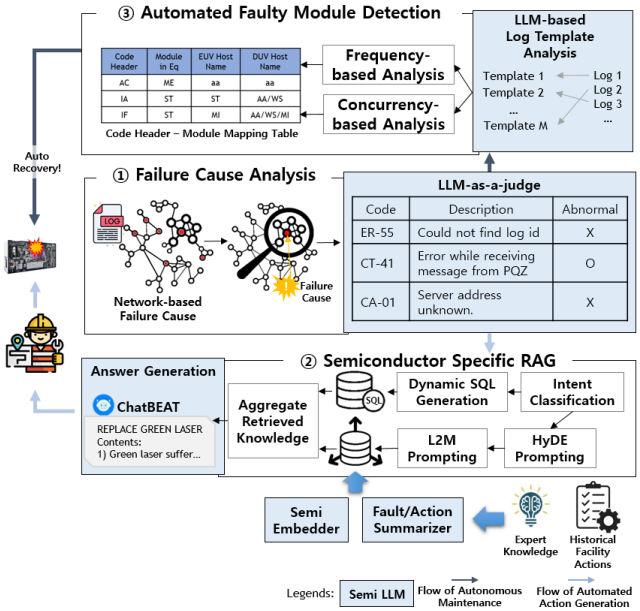

Figure 1: Overview of the deployed autonomous maintenance system. The pipeline integrates failure detection, graph-based inference, RAG-based action planning, and continuous feedback reflection in a closed-loop workflow.

- Failure cause inference module combining event propagation graphs with LLM-based semantic ranking: This module overcomes the limitations of traditional time-series data-based failure cause diagnosis, enabling more accurate diagnosis.
- Domain-adapted RAG pipeline: The proposed RAG pipeline addresses the limitations of human-dependent action generation, leveraging symbolic SQL retrieval and hybrid prompting to enable efficient and accurate action generation.
- Deployed fab at Samsung Electronics, the system reduced MTTR by 7 minutes on average, and autonomously resolved over 70% of software-level failures.

This work illustrates how tightly integrated reasoning pipelines enable reliable autonomous recovery in real-world semiconductor fabs.

2 Related Work

2.1 Failure Diagnosis in Semiconductor Manufacturing

Failure cause analysis is critical in semiconductor manufacturing for identifying failure causes and preventing recurrence. This process involves the analysis of sensor data such as temperature, pressure, and vibration, as well as event logs, to monitor equipment conditions and identify failure patterns. Traditionally, failure cause analysis often rely on time series data collected from various sensors in the manufacturing environment. These sensors provide crucial data points that, when analyzed, can reveal abnormal conditions indicative of potential failures. Various machine learning models for fault detection and classification (FDC) have been employed to detect anomalies and infer failure causes in semiconductor manufacturing processes. However, these approaches have limitations, such as the need to identify which sensors to monitor for each case of equipment failure and the inability to fully utilize text-based event logs that provide comprehensive information about the actual equipment status [3, 4]. Also, failure diagnosis in semiconductor manufacturing has been predominantly expert-driven. Engineers utilize their experience and knowledge to interpret sensor data and event logs, infer causes, and decide on corrective actions. This approach, while valuable, is inherently limited by the availability and expertise of the personnel involved [7, 15].

Event logs capture real-time equipment status and activities and are essential for failure cause analysis. However, the cause has traditionally been identified in a rule-based manner, relying heavily on the expertise of engineers. While the inverse frequency approach proposed by Ki et al. [10] marked an important first step in leveraging event logs for failure cause inference, its effectiveness is fundamentally constrained by two key limitations. First, frequency-based heuristics assume that rare events are inherently abnormal, which often fails in scenarios where critical failures arise from common but contextually significant events. Second, this method lacks structural awareness of causal dependencies between events and cannot account for the temporal or relational context within complex tool sequences. As a result, such methods struggle to generalize across diverse failure modes or prioritize the most plausible failure causes in ambiguous multi-event scenarios.

To address these limitations, a more robust diagnostic system must integrate topological patterns from event graphs, historical engineer feedback, and domain-grounded semantic reasoning. Our proposed system fills this gap by combining centrality-based network analysis with LLM-based scoring and expert-guided filtering to deliver highly accurate and explainable failure cause predictions in production-scale environments.

2.2 Domain-Specific Adaptation of Large Language Models

General-purpose large language models (LLMs) have achieved impressive results across a variety of tasks, but their application to highly specialized domains such as semiconductor manufacturing remains challenging. This is largely due to a mismatch in vocabulary, domain knowledge, and contextual semantics. In particular, semiconductor maintenance scenarios involve highly specialized terminology, numeric event codes, and process-specific logic that are poorly represented in general training corpora.

There are two critical limitations when deploying off-the-shelf LLMs in this setting. First, they often fail to correctly interpret or reason over specialized terms and structured identifiers that are central to maintenance logs and engineering procedures. Second, they lack the contextual grounding needed to understand the procedural relationships between failures, diagnostics, and corrective actions in complex manufacturing systems.

Prior research in other domains has demonstrated the effectiveness of domain-specific adaptation. Models such as BioBERT [11] and ClinicalBERT [1] achieved significant improvements over general LLMs by continuing pretraining on domain-specific corpora.

Gururangan et al. [8] further emphasized the importance of ongoing pretraining and task-specific fine-tuning to adapt LLMs to specialized use cases.

Inspired by these findings, our system adapts LLMs to the semiconductor maintenance domain by incorporating vocabulary from historical equipment logs, engineering reports, and failure-action documents. This adaptation enhances the LLM's ability to reason over domain-specific inputs and generate accurate, contextually appropriate maintenance actions. As demonstrated in our experiments, the domain-adapted LLM significantly outperforms generic models in both retrieval-augmented generation and autonomous reasoning tasks under real-world fab conditions.

Beyond model adaptation, retrieval strategies must also be tailored to semiconductor-specific knowledge and language. Recent advancements in Retrieval-Augmented Generation (RAG) focus on enhancing the synergy between retrieval mechanisms and large language models to improve performance on complex tasks. Techniques such as Interleaving Retrieval with Chain-of-Thought Reasoning [16], Step-Back Prompting [17], Query Rewriting [12], Hypothetical Document Embeddings [6], and Least-to-Most Prompting [18] have been introduced to improve retrieval relevance, reasoning depth, and adaptability to complex queries. While these advancements have improved general reasoning and retrieval alignment in open-domain tasks, applying RAG to high-precision industrial domains like semiconductor maintenance poses unique challenges, including symbolic schema grounding, real-time constraints, and engineering-specific terminology. General-purpose retrievers often fail to match domain-specific queries involving numerical event codes or ambiguous maintenance phrasing. To address this, we build a domain-specific RAG pipeline that leverages fine-tuned semantic embedding models, hybrid prompting strategy, and pre-annotated fault-action pairs for grounding. This enables accurate retrieval and contextual augmentation for LLMs, significantly improving reasoning precision in fab environments.

3 Multi-Stage Reasoning Pipeline for Fab-Level Autonomous Recovery

3.1 Failure Cause Analysis with Network, Feedback, and LLM Reasoning

To determine the most likely cause of equipment failures, our system constructs a directed event-event graph from recent logs and performs a multi-stage analysis that combines graph topology, historical feedback, and semantic reasoning.

3.1.1 Event Log Graph Construction. Given a chronological sequence of event codes $\mathcal{L} = [e_1, e_2, \ldots, e_T]$ extracted from equipment logs, a directed graph $G = (V, E)$ is constructed, where:

- V is the set of unique event codes observed.
- $(v_i, v_j) \in E$ if event v_i precedes v_j within a fixed time window Δt with conditional probability $P(v_j|v_i) > \theta$.

The resulting graph captures temporal propagation between alarms and process anomalies.

3.1.2 Betweenness Centrality-Based Candidate Ranking. To identify influential nodes in failure propagation, the **betweenness centrality** $C_B(v)$ of each node v is computed as:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{1}$$

where σ_{st} is the number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of such paths passing through v [5].

Nodes with high $C_B(v)$ are selected as initial failure cause candidates. However, structural centrality alone is insufficient to reflect domain correctness.

3.1.3 Feedback-Based Reweighting. To incorporate field expertise, the centrality scores is adjusted based on historical engineer feedback. For each node v, we maintain:

- $f^+(v)$: number of times v was confirmed as true cause.
- $f^-(v)$: number of times v was ruled out.

A feedback weight is computed as:

$$w_{\text{fb}}(v) = \frac{f^+(v)}{f^+(v) + f^-(v) + \epsilon}, \tag{2}$$

where ϵ is a smoothing constant. The feedback-adjusted score is:

$$C'_B(v) = C_B(v) \cdot w_{\text{fb}}(v). \tag{3}$$

This process adapts centrality ranking over time to reflect real-world misdiagnoses and corrections.

3.1.4 LLM-Based Semantic Scoring. Refined candidates are further evaluated using a domain-specialized LLM to estimate how strongly each event description indicates a hardware malfunction. Instead of relying on symbolic codes, we use the full natural language descriptions associated with each event (e.g., "Chamber vacuum pressure abnormal", "Stepper motor misalignment detected").
The LLM receives the top-k event descriptions in plain text and assigns a malfunction likelihood score $s_{\text{LLM}}(v) \in [0, 1]$ to each.
The model output is a mapping:

$$\{v_i \rightarrow s_{\text{LLM}}(v_i)\}_{i=1}^{k} \tag{4}$$

These scores are then combined with the feedback-adjusted centrality score $C'_B(v)$ to compute the final ranking score:

$$S(v) = \lambda \cdot C'_B(v) + (1 - \lambda) \cdot s_{\text{LLM}}(v) \tag{5}$$

This design allows the system to distinguish events that are semantically associated with hardware failures from those caused by software, recipe, or user-level issues, even when the network topology is ambiguous.

3.1.5 Advantages of Hybrid Analysis. The multi-stage hybrid diagnosis framework combines the strengths of statistical structure, user-driven adaptation, and language-based reasoning. Each layer addresses distinct failure modes:

- **Structural grounding (Centrality)**: Betweenness centrality captures typical fault propagation pathways and provides a fast, interpretable first-pass candidate ranking.
- **Operational adaptivity (Feedback)**: Engineer feedback reweights misleading centrality results, especially for evolving equipment behaviors and non-causal alarm noise.

- **Contextual reasoning (LLM)**: The language model scores each event's plausibility based on semantic similarity to historical descriptions of true hardware faults. This step corrects for false positives in both graph topology and misclassified feedback.

This modular integration improves resilience to edge conditions such as:

- *Ambiguous log sequences* (e.g., frequent non-causal alarms),
- *Novel fault types* not seen in training,
- *Tool-specific anomalies* where symbolic heuristics are brittle.

Beyond top-1 accuracy (98.6% over 3,812 failure cases), the system contributed 7 minutes average reduction in overall Mean Time To Recovery(MTTR). These results demonstrate that hybridization is not merely stacking methods, but a failure-tolerant, domain-aligned design pattern for real-world deployment.

3.2 Semiconductor-Specific RAG Pipeline

Once the failure cause is identified through the hybrid graph and LLM reasoning pipeline described in Section 3.1, the system initiates a follow-up process to recommend appropriate recovery actions. Instead of relying on static rule mappings, the action recommendation module dynamically constructs a natural language query based on the diagnosed cause. This cause-aware query is then passed into a semiconductor-specific Retrieval-Augmented Generation (RAG) pipeline, which retrieves relevant documents and synthesizes actionable recovery plans.

The pipeline combines domain-tuned embeddings, intent classification, symbolic and neural retrieval, and hybrid prompting to ensure high-precision action generation.

3.2.1 Understanding Query Intent with Domain-Specific Preprocessing. To interpret cause-aware queries or operator utterances, the system applies a domain-specialized intent analysis module:

- **Domain-Specific NER**: A fine-tuned Named Entity Recognition module extracts key equipment-specific concepts—such as module names, sensor types, and process steps—from each query.
- **Intent Classification**: Based on the extracted entities, the system uses a rule-based mapping table—constructed through manual analysis of real-world engineer queries—to classify each query into either (1) `structured retrieval` or (2) `semantic retrieval`.

This rule-based intent mapping enables robust and interpretable routing, without relying on opaque classification models.

3.2.2 Structured Retrieval with LLM Generated Dynamic Query. For queries classified as structured—typically those referencing both a module and a alarm code—the system activates a **LLM-in-the-loop symbolic retriever**. This component transforms natural language into executable SQL for precise access to structured knowledge bases, such as fault-action tables or incident logs.

Instead of relying on static templates, the retriever uses prompt-engineered instructions to elicit SQL queries from a domain-adapted LLM. The prompt includes:

- A natural language query reformulated from the failure cause or user input,
- The schema of the target table (e.g., ActionKB(Module, Symptom, Action)),
- Example records or field types for grounding.

Example Prompt:

```
Task: Write an SQL query to retrieve recommended
actions.
Schema: ActionKB(Module TEXT, Symptom TEXT,
Action TEXT)
Query: What should I do when the bake module
temperature rises above 200C?
Output: SELECT Action FROM ActionKB WHERE
Module = 'Bake' AND Symptom LIKE '%temperature%'
```

The LLM interprets context and generates executable queries, which are then issued to internal databases. This process supports:

- Precise retrieval when structured metadata is available,
- Adaptability to schema evolution without hard-coded templates,
- Explainability via symbolic traces.

3.2.3 Semantic Search with Fine-Tuned Embedding Models and Hybrid Prompt Strategies. To enable robust similarity matching between queries and recovery documents, we fine-tuned a base `bge-m3` embedding model on a domain-specific corpus of over 50,000 failure-action triplets. Each training instance consists of:

- an anchor query q derived from a real user question or diagnosed failure cause,
- a positive document d^+ describing a successful solution,
- a negative document d^- sampled from unrelated actions or failure categories.

We used triplet loss to optimize the embedding space such that:

$$\cos(\phi(q), \phi(d^+)) > \cos(\phi(q), \phi(d^-)) + \delta \tag{6}$$

where δ is a margin parameter. This objective encourages the model to map semantically related failure-action pairs closer together. Figure 2 shows a t-SNE visualization of the embedding space before and after fine-tuning, where fine-tuned embeddings exhibit tighter clustering of documents by failure category.

The fine-tuned bge-m3 model captures semiconductor-specific terminology and subtle distinctions in failure cases. This adaptation improved top-1 match accuracy on the BMQA dataset from 0.639 to 0.709 (Section 4.2), enabling retrieval of rare but critical actions.

To handle ambiguous, vague, or multi-hop queries, the system employs a hybrid prompting strategy that integrates **Hypothetical Document Embedding (HyDE)** and **Least-to-Most (L2M)** prompting. These two methods operate sequentially to enable robust reasoning in data-sparse or linguistically underspecified scenarios.

Step 1: HyDE – Hypothetical Document Embedding. The system prompts the LLM to hallucinate a plausible incident description based on the query. This pseudo-document expands retrieval coverage and guides document selection.

Step 2: L2M – Least-to-Most Prompting. The retrieved documents and original query are then passed to an LLM using L2M prompting to break the problem into tractable substeps and reason through them incrementally.

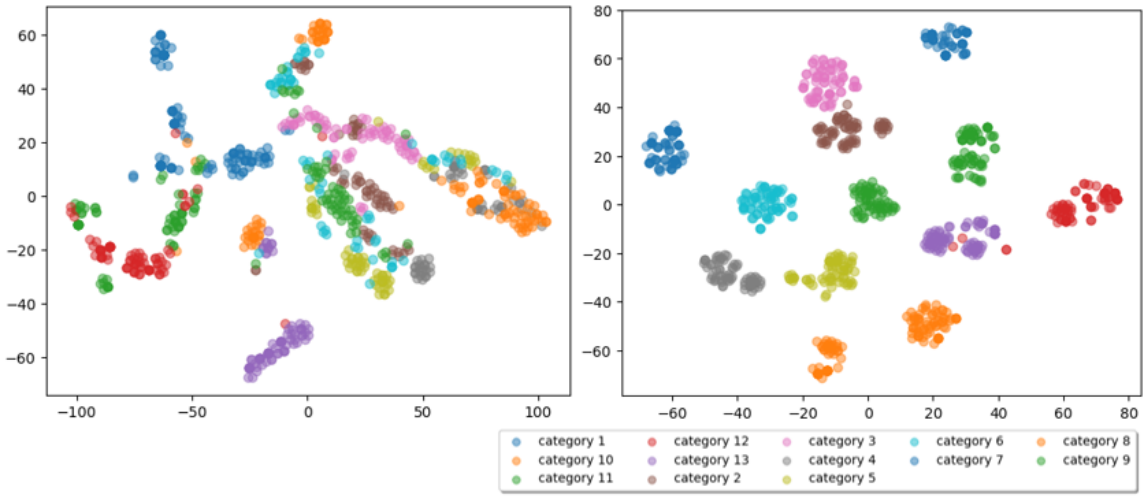

Figure 2: t-SNE visualization: base bge (left) vs fine-tuned (right).

This pipeline enables context-sensitive, grounded, and explainable recovery guidance across both symbolic and neural pathways.

3.3 Automated Recovery Execution via Control Software Maintenance

While Section 3.2 addressed human-in-the-loop recommendations for hardware failures, this section describes a fully automated recovery pathway that directly interfaces with equipment control software. By automating procedures for 70% of breakdowns, the system reduces MTTR and minimizes manual intervention.

3.3.1 LLM-Based Event Template Analysis. When a failure cause is identified (Section 3.1), the system prompts a domain-adapted LLM to convert the raw event log into a canonical failure template. Given the variability of log formats, this normalization step ensures consistency across equipment models and vendors.

3.3.2 Module Mapping via Engineering Knowledge Base. The system then identifies the affected module by referencing a curated mapping table that links canonical templates to modules. This table is built from expert knowledge, operational history, and continuously refined via feedback.

3.3.3 Control Software-Driven Execution. With the target module identified, the system triggers a predefined recovery sequence through the control software:

- Engineers are notified via popup to avoid manual conflict.
- Pre-maintenance tasks (e.g., config backup, log archiving) are auto-triggered.
- Recovery actions (e.g., reboot, parameter reset) are executed via secure APIs.
- Post-checks verify success by confirming alarm clearance and sensor stability.

This pipeline enables fast, consistent, and conflict-free recovery in diverse fab environments.

3.4 Knowledge Grounding and Feedback Loop for Reliable Action Recommendation

To support robust and adaptive recovery reasoning, we introduce a knowledge structuring and refinement module that continuously curates domain knowledge and integrates field feedback.

3.4.1 Knowledge Extraction and Structuring. Since January 2021, we have collected a large corpus of fab operation documents, including alarm logs, maintenance reports, and engineering notes, covering more than four years of field incidents and accumulating over 220,000 documents. By analyzing the templates of these documents, we extracted failure causes and recovery actions written by engineers, resulting in structured data pairs (failure cause, recovery action), which helps to avoid hallucination when using these unstructured sources for retrieval-augmented generation (RAG).

The extracted pairs are stored in an Action Knowledge Base (ActionKB), which serves as a statistically grounded reference. For instance, when a specific equipment alarm occurs, the ActionKB may indicate that "part replacement" was the recovery action taken in 80% of similar past incidents, with "software update" being the cause in 60% of those cases. The system uses this KB not only to provide direct recommendations, but also to show frequency-based statistics that guide engineers toward empirically validated solutions.

3.4.2 Context Expansion via Vector Search. For complex or ambiguous queries, the system complements ActionKB lookups with retrieval from a continuously updated vector database. This knowledge source includes expert notes and daily maintenance updates, enabling dynamic and context-aware generation of troubleshooting workflows, adjustment guides, and failure advisories.

3.4.3 Feedback-Driven Refinement. To improve accuracy and trust over time, the system supports feedback from engineers on both failure cause inference and action recommendations. These signals are used to re-rank candidates in the failure cause graph (Section 3.1) and adjust the reliability scoring in ActionKB.

Through this loop, the system evolves with real-world patterns, providing future users with more reliable and empirically grounded decisions.

4 Empirical Evaluation in Semiconductor Fabs

We evaluated our system using over 3,800 real-world equipment breakdown cases collected from 762 photolithography equipment at Samsung Electronics fab between January 2021 and April 2025. This evaluation focuses on two core objectives: (1) failure cause identification using event-log-based graph analysis, and (2) recovery action generation accuracy based on a domain-specific QA benchmark. Together, these results demonstrate the system's effectiveness not only in controlled experiments but also in operational fab settings.

4.1 Failure Cause Inference from Event Logs

To evaluate the performance of our failure cause inference module (Section 3.1), we conducted a comprehensive empirical study using 3,812 real breakdown cases. These incidents span four years and cover diverse failure types encountered across 762 photolithography equipment. Unlike synthetic benchmarks, this evaluation adheres to the standards and practices of semiconductor fab engineers, providing insights into production-level robustness.

For each incident, the top-ranked system-inferred root cause was compared against the ground-truth diagnosis annotated by field engineers. The evaluation metric used was the top-1 match rate between system predictions and expert labels.

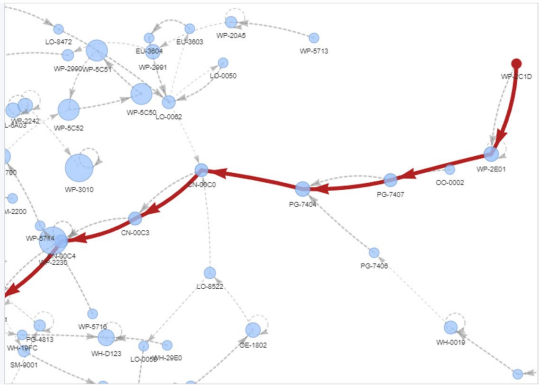

Figure 3: Event propagation graph visualization from real fab logs.

As summarized in Table 1, baseline methods such as autoencoder-based anomaly detection (0.614) and polarity scoring (0.628) showed limited performance due to their lack of structural context awareness. Each of the two methods was inspired by research on Heerschop et al.[9] and Cheng et al. [2]. Graph-based methods significantly improved accuracy by incorporating propagation structure; specifically, using betweenness centrality alone yielded 0.782, while adding reweighting based on historical feedback improved it to 0.892. Our proposed method, combining graph-based centrality with semantic scoring from a large language model, achieved the highest accuracy at 0.986. This demonstrates that LLM-based contextual disambiguation is particularly effective in distinguishing between ambiguous or overlapping event patterns. As shown in Figure 3, the system learns to model causal chains within complex equipment logs, allowing it to correctly identify dominant fault signatures.

Method	Top-1 Match Rate
Autoencoder-Based Anomaly Detection	0.614
Text Polarity Scoring	0.628
Graph Centrality Only	0.782
+ Feedback Reweighting	0.892
+ LLM Semantic Scoring	**0.986**

Table 1: Failure cause identification accuracy across 3,812 breakdowns. Our system includes Graph Centrality Only, Centrality + Feedback Reweighting, and Centrality + LLM Semantic Scoring.

In addition to accuracy, we validated system robustness at scale. During the deployment period, our system was used in live production across all 762 equipment, analyzing a total of 70,961 breakdown events. Each case was automatically diagnosed and prioritized within 1 minute of occurrence. The system's ability to operate under high throughput conditions while maintaining precision and latency requirements confirms its practicality for real-world fab operations. Though this evaluation focuses on photolithography equipment, the architecture is generalizable to other modules such as etchers or cleaners that follow similar event structures.

4.2 Evaluation of Action Recommendation with BMQA

We next assessed the effectiveness of our action recommendation module described in Section 3.2. For this, we used the Breakdown Maintenance QA (BMQA) benchmark—a domain-specific dataset consisting of 6,845 QA pairs that map real failure causes to validated corrective actions. These pairs were derived from real incident reports and reviewed by experienced fab engineers, making the benchmark reflective of realistic operational conditions.

To evaluate recommendation quality, we used an embedding similarity threshold to determine match correctness: if the cosine similarity between the predicted and reference answer embeddings exceeded 0.7, the prediction was considered correct. The BMQA score thus reflects the percentage of semantically aligned responses across all QA pairs.

$$\text{BMQA}_{0.7} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[\text{sim}(\hat{a}_i, a_i) \geq 0.7\right] \tag{7}$$

As shown in Table 2, our evaluation proceeded in an ablation-style fashion to isolate the contribution of each component. The baseline BM25 retriever achieved 0.601, while semantic RAG (using dense retrieval only) improved this to 0.639. Introducing domain-specific embedding fine-tuning resulted in 0.709 by enhancing the model's sensitivity to context-specific fault terms (e.g., distinguishing "chamber overpressure" from "pump underflow"). Adding intent classification boosted performance to 0.755 by accurately routing queries to either symbolic or generative submodules, particularly

Method Configuration	BMQA@0.7
BM25-Based Retrieval	0.601
Semantic RAG (dense retriever only)	0.639
+ Domain-Tuned Embeddings	0.709
+ Intent Classification	0.755
+ Hybrid Prompting (HyDE + L2M)	0.846
+ Symbolic SQL Retrieval (Full System)	**0.862**

Table 2: BMQA evaluation across retrieval configurations.

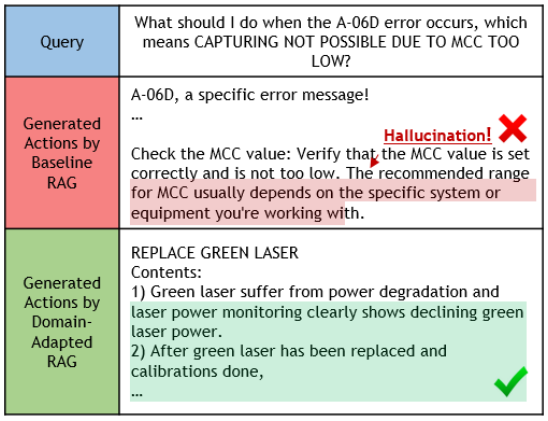

Figure 4: Comparison between baseline RAG and domain-adapted RAG.

benefiting structured queries with numerical constraints. Further gains were achieved by integrating hybrid prompting, combining HyDE-based hallucinated contexts with Least-to-Most reasoning for step-wise decomposition. This setup improved interpretability and coverage, yielding 0.846. Finally, incorporating symbolic SQL retrieval pushed performance to 0.862. This component proved highly effective in repetitive, high-frequency failure scenarios, enabling fast and deterministic lookups of known (module, symptom) combinations.

Figure 4 illustrates a comparison between our domain-adapted RAG and the baseline RAG utilizing Llama4-maverick as the LLM. The comparison highlighted that the baseline RAG, which is based on semantic search, often resulted in hallucinations. In contrast, our domain-adapted RAG, trained with domain-specific data, did not exhibit such hallucinations and demonstrated superior accuracy and relevance in its recommendations.

4.3 Real-World Validation of Autonomous Maintenance Execution

To validate the full-stack deployment of our autonomous maintenance system—including failure inference, action recommendation, and automated control execution—we analyzed its operational performance from January 2021 to April 2025 across 762 semiconductor equipment.

Among the 3,812 breakdown cases recorded during this period, 72.3% were categorized as short-term recoverable failures such as initialization errors, pressure threshold violations, or robot alignment issues. Of these, 2,743 incidents were fully resolved via automated recovery scripts triggered by the system. These actions required no human intervention aside from a confirmation click in the user interface, demonstrating the system's ability to handle frequent disruptions autonomously.

The deployment also led to a marked reduction in MTTR. While manual recovery averaged 9.2 minutes, automated handling reduced this to 2.1 minutes on average, with best-case latencies under 90 seconds. Over the full deployment span, this yielded an average reduction of 7 minutes per breakdown.

To prevent coordination issues between automated and manual actions, which initially caused conflicts, we introduced a real-time alert system in the equipment UI. This notifies engineers when the system has taken autonomous action, ensuring operational transparency. Since its rollout in March 2025, there have been no cases of misoperation or conflicting intervention, maintaining a 100% success rate in automated recoveries.

Beyond metrics, this deployment has shifted the maintenance paradigm. By offloading repetitive diagnostics and recovery tasks, engineers can now focus on complex process optimization and long-term analysis. Our system's reliable performance under production conditions validates its role as a foundation for truly unmanned semiconductor maintenance workflows.

5 Conclusion

We presented a production-grade autonomous maintenance system for semiconductor manufacturing, integrating event log graph analysis, LLM-based failure reasoning, and domain-adapted retrieval-augmented generation (RAG). The system identifies failure causes using betweenness centrality on event graphs, abstracts log templates via LLM prompting, and recommends recovery actions through a hybrid pipeline combining symbolic SQL retrieval and dense document search, grounded in historical fab knowledge.

Deployed across 762 photolithography equipment at Samsung Electronics fab, the system analyzed over 3,800 real breakdowns and autonomously resolved 72.3% of short-term failures, reducing average MTTR by 7 minutes. Following the addition of real-time UI alerts, autonomous execution achieved a 100% success rate without operator conflict. On the BMQA benchmark, the system improved answer similarity by +0.261 over semantic RAG, confirming its effectiveness in retrieval and generation under fab-specific constraints.

As the industry moves beyond breakdown maintenance, the importance of predictive maintenance research is increasingly emphasized. Recent studies highlight the growing need for proactive maintenance strategies [13, 14]. By adopting predictive maintenance, organizations can reduce downtime and improve overall efficiency. Further investigation into predictive maintenance is crucial for advancing this field.

This work demonstrates that combining reasoning and retrieval with domain-specific adaptation can enable robust, real-time automation in high-stakes industrial environments. By closing the loop between observation, interpretation, and action, our system advances the practical deployment of LLMs in mission-critical manufacturing contexts.

6 GenAI Usage Disclosure

Generative AI tools (Llama 4 Maverick-based in-house internal AI Assistant) were used primarily for translation and to improve readability of the manuscript. All content, ideas, and conclusions are the authors' own, and the authors take full responsibility for the paper's content.

References

[1] Emily Alsentzer, John R Murphy, Willie Boag, Wei-Hung Weng, Di Jin, Tristan Naumann, and Matthew McDermott. 2019. Publicly available clinical BERT embeddings. *arXiv preprint arXiv:1904.03323* (2019).

[2] Zhen Cheng, En Zhu, Siqi Wang, Pei Zhang, and Wang Li. 2021. Unsupervised outlier detection via transformation invariant autoencoder. *IEEE Access* 9 (2021), 43991–44002.

[3] Mustafa Demetgul. 2013. Fault diagnosis on production systems with support vector machine and decision trees algorithms. *The International Journal of Advanced Manufacturing Technology* 67 (2013), 2183–2194.

[4] Shu-Kai S Fan, Chia-Yu Hsu, Du-Ming Tsai, Fei He, and Chun-Chung Cheng. 2020. Data-driven approach for fault detection and diagnostic in semiconductor manufacturing. *IEEE Transactions on Automation Science and Engineering* 17, 4 (2020), 1925–1936.

[5] LC Freeman. 1977. A set of measures of centrality based on betweenness. *Sociometry* (1977).

[6] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1762–1777.

[7] Crina Grosan, Ajith Abraham, Crina Grosan, and Ajith Abraham. 2011. Rule-based expert systems. *Intelligent systems: A modern approach* (2011), 149–185.

[8] Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. *arXiv preprint arXiv:2004.10964* (2020).

[9] Bas Heerschop, Frank Goossen, Alexander Hogenboom, Flavius Frasincar, Uzay Kaymak, and Franciska de Jong. 2011. Polarity analysis of texts using discourse structure. In *Proceedings of the 20th ACM international conference on Information and knowledge management*. 1061–1070.

[10] Wanwook Ki, Minjae Choi, Yongjun Lee, Jisoo Park, Ji Jiyeon Park, Jaeyong Park, and Younghoon Kim. 2024. Predictive Analysis and Root Cause Identification of Equipment Failures Using Event Log Patterns: A Focus on EUV/DUV Equipment. In *2024 35th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC)*. IEEE, 01–05.

[11] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (2020), 1234–1240.

[12] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. 2023. Query rewriting in retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 5303–5315.

[13] Sathyan Munirathinam and Balakrishnan Ramadoss. 2014. Big data predictive analtyics for proactive semiconductor equipment maintenance. In *2014 IEEE International Conference on Big Data (Big Data)*. IEEE, 893–902.

[14] Anis Ben Said, Muhammad-Kashif Shahzad, Eric Zamai, Stephane Hubac, and Michel Tollenaere. 2016. Towards proactive maintenance actions scheduling in the Semiconductor Industry (SI) using Bayesian approach. *IFAC-PapersOnLine* 49, 12 (2016), 544–549.

[15] Afolalu Sunday, Salawu Enesi, Oluyemi Kehinde, Ayuba Samuel, Ihebom Ikechi, and Elewa Remilekun. 2018. Failure mode and effect analysis a tool for reliability evaluation. *European Journal of Engineering Research and Science* 3, 4 (2018), 65–68.

[16] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. *arXiv preprint arXiv:2212.10509* (2022).

[17] Huaixiu Steven Zheng, Swaroop Mishra, Xinyun Chen, Heng-Tze Cheng, Ed H Chi, Quoc V Le, and Denny Zhou. 2023. Take a step back: Evoking reasoning via abstraction in large language models. *arXiv preprint arXiv:2310.06117* (2023).

[18] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, et al. 2022. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625* (2022).